

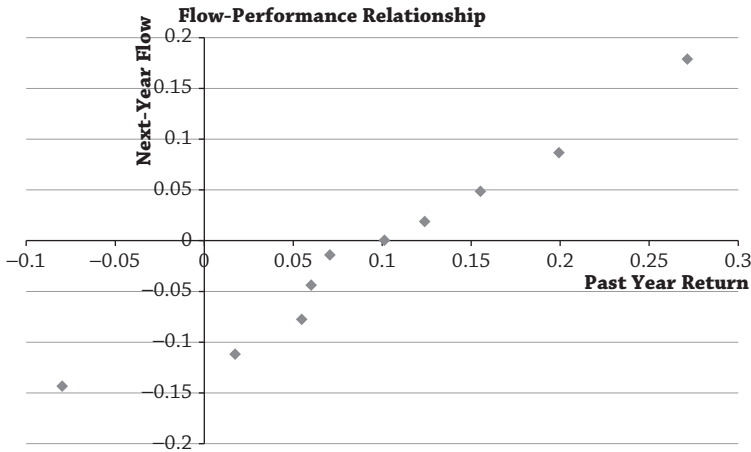


Figure 16.5

a paper in 2004 that fits the main stylized facts of mutual fund performance: (i) managers add value on a before-fee basis but underperform, on average, after fees; (ii) investors chase returns; and (iii) investors end up doing poorly when they move money into funds with the highest past returns. In the field of delegated asset management, this is the most important paper published in the past thirty years.

Berk and Green (2004) show that these facts result from a *rational equilibrium* in which fund managers have talent, on average, but some investors are talented and some are not (there is *differential ability* to generate excess returns). Investors know that some managers have talent. Fund managers, however, have *decreasing returns to scale*: as their funds become bigger, the excess returns they can generate shrink as we observe in data.

Investors know who the skilled managers are, so money flows to the best managers first. These are the managers that have generated past high returns—some of this is luck, but some of it is because they truly are talented. This explains why investors chase returns. As money flows in, the funds get bigger. But as these funds increase in size, the managers' alphas disappear. Thus there is little persistence in performance.

In allocating money, asset owners go with the best manager. The new money reduces her ability to generate high returns and drives her expected return down to the second best manager's return. At that point, asset owners put money with the second best manager. The same thing happens until the expected returns of the first two managers are forced down to equal the expected return of the third manager, and so on. This continues until all investors are indifferent between investing actively or just putting their money in a cheap, passive index fund. Although fund managers have talent, investors receive returns that are the same as the market. Tack on fees, and active mutual fund returns underperform the market.